

- Q.27 Write a short note on Outdoor Substation?
- Q.28 Write any four advantages of H.V.A.C. transmission system?
- Q.29 Define Tariff and what are its objective?
- Q.30 Discuss any four advantages and disadvantages of Steam Power Station.
- Q.31 Define the following:
  - i) Geothermal energy
  - ii) Solar energy
- Q.32 Explain Ring main system.
- Q.33 Write all the factors that affects the Corona loss.
- Q.34 Draw and explain Steel Tower.
- Q.35 Why voltage is stepped down at Substation?

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write a short note on:
- a) Merits & demerits of Diesel Power Plant.
  - b) Merits & demerits of steam Power Plant
- Q.37 Draw & discuss the layout of 3 phase transmission and distribution system.
- Q.38 Compare overhead Distribution system versus under ground Distribution system.

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**5th Semester/ Elect., Elect & Eltx Engg.**  
**Subject : Electrical Power-I/Power-I (G.T.&D.E.P.)**

Time : 3 Hrs.

M.M. : 100

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which support for overhead transmission lines has the least life?
- a) Wooden poles
  - b) Fabricated steel structure
  - c) R.C.C. Poles
  - d) Steel poles
- Q.2 A Load curve is a plot of
- a) Load v/s time
  - b) Load v/s current
  - c) Load v/s voltage
  - d) Total no of units generated v/s time
- Q.3 Which distribution system is more reliable?
- a) Radial
  - b) Tree System
  - c) Ring Main
  - d) All of these
- Q.4 The power factor of an inductive circuit is
- a) Leading power factor
  - b) Lagging power factor

- c) Unity power factor
  - d) None of these
- Q.5 Annual operating cost of generating plant consists of
- a) Fixed charges                      b) Semi-fixed charges
  - c) Operating charges              d) All of these
- Q.6 Primary distribution lines are
- a) Reactors
  - b) Feeders
  - c) Sub-transmission lines
  - d) None of these
- Q.7 HVDC stands for
- a) Corona loss
  - b) High voltage direct current
  - c) High value direct current
  - d) High voltage alternating current
- Q.8 Which of the following is not a renewable energy source?
- a) Nuclear                              b) Wind
  - c) Sun                                      d) Tidal energy
- Q.9 Power is distributed by a system called
- a) DC, 2 wire system
  - b) AC, 3 phase, 3 wire system
  - c) AC, 2 wire system
  - d) AC, 3 phase, 4 wire system
- Q.10 Sag is inversely proportion to the
- a) Weight of conductor      b) Tensile strength
  - c) Length                              d) All of these

## SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 The maximum value of power factor is \_\_\_\_\_.
- Q.12 OTEC stands for \_\_\_\_\_.
- Q.13 A diesel power plant is best suited as stand by plant. (True/False)
- Q.14 A.C.S.R.stands for \_\_\_\_\_.
- Q.15 Diversity is the ratio of \_\_\_\_\_ to system maximum demand.
- Q.16 The cost of indoor substation is \_\_\_\_ than the outdoor substation of same capacity.
- Q.17 Reactive power unit is \_\_\_\_\_.
- Q.18 Wind pressure acts \_\_\_\_ to the weight of conductor.
- Q.19 The main source of energy is \_\_\_\_\_.
- Q.20 \_\_\_\_\_ is the main waste product of the steam plant.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Name & Explain different types of insulators used for transmission lines.
- Q.22 Define and explain String Efficiency.
- Q.23 Define the following terms:
  - a) Demand factor              b) Load curve
- Q.24 How can Power factor be improved?
- Q.25 What do you mean by underground distribution system?
- Q.26 Mention merits of Ring Distribution over Radial distribution system.